
MACHINE LEARNING

WORKSHOP

AUTHORED BY

GUSTAV SØNDERGAARD NYBRO
NIKOLAJ ALBERTSEN
MAGNUS VØRS HOLMSGÅRD
RUI MARIA MARTINS LOUREIRO
BJARKI FRÓÐASON Í EYÐANSSTOVU

AALBORG UNIVERSITY

$\mathcal{PL}$

2026

GROUP 630

1 Lecture 2

Hand written Exercise 1

A bank wants to classify loan applicants (x) as either low-risk (C_1) or high-risk (C_2). For a new applicant, a Gaussian classifier gives the following likelihoods:

$$p(x|C_1) = 0.04$$

$$p(x|C_2) = 0.07$$

From historical data, the class priors are:

$$p(C_1) = 0.8$$

$$p(C_2) = 0.2$$

- Compute the unnormalized posterior for each class.
- Classify the applicant using the unnormalized posterior.
- Repeat the classification assuming uniform priors.
- Explain why the decision changes or does not change.

Solutions

- a) When it is unnormalized, we do not have to divide by $P(x)$, so the unnormalized posterior is given by: Likelihood * Prior.

$$C1 : P(C1|x) = P(x|C1) * P(C1) = 0.04 * 0.8 = 0.032$$

$$C2 : P(C2|x) = P(x|C2) * P(C2) = 0.07 * 0.2 = 0.014$$

- b) Since

$$P(C1|x) > P(C2|x)$$

the applicant is classified as low-risk.

- c) (Uniform priors)

$$P(C1) = P(C2) = 0.5$$

$$P(C1|x) = P(x|C1) * P(C1) = 0.04 * 0.5 = 0.02$$

$$P(C2|x) = P(x|C2) * P(C2) = 0.07 * 0.5 = 0.035$$

Since

$$P(C2|x) > P(C1|x)$$

the applicant is classified high risk.

- d) Uniform priors remove historical class prevalence from the equation, reducing the classification rule purely to maximum likelihood estimation. When historical priors are included, the strong prior belief the most applicants are low-risk heavily counteracts the higher likelihood of class C2, shifting the overall decision back to C1.

Hand Written Exercise 2

Consider the following two-dimensional training data belonging to two classes:

Point	Coordinate	Class
A	(1,1)	C_1
B	(2,2)	C_1
C	(3,2)	C_2
D	(5,4)	C_2
E	(4,1)	C_2

Table 1: Caption

A test sample, F, is observed at: $F = (3, 1)$

- Compute the Euclidean distance from the test point to each of the five training points.
- Rank the training points from nearest to farthest.
- Classify the test point using 1-nearest neighbor (1-NN).
- Classify the test point using 3-nearest neighbors (3-NN).
- Classify the test point using 5-nearest neighbors (5-NN).

Solutions

- a) To compute the euclidean distance between the observed point F and training data points, we use:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

giving us the distances:

$$d_A = 2$$

$$d_B = 1.414$$

$$d_C = 1$$

$$d_D = 3.6$$

$$d_E = 1$$

- b) The points ranked from nearest to farthest are:

C

E

B

A

D

- c) For $k = 1$ the observed point would belong to the C_2 class
d) For $k = 3$ the observed point would belong to the C_2 class

- e) When $k = 5$ we include all the training data, and the combined distance from class C_1 is smaller, that is the assigned class of the observed point.